Module 1 Lab Packet — Data Types, Expressions & Flow Control (Java)

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How to use this packet

This packet holds every hands-on exercise for data types, expressions and flow control. Work in your `claim-academy-[name]-sdai-2026-cohort` repo and commit after each exercise.

#	Exercise	Mode	Deliverable
1	Division & casting warm-up	Solo, by hand then in code	`DivisionWarmup.java`
2	Grade Calculator	Guided by instructor	`GradeCalculator.java`
3	Number Game	Independent	`NumberGame.java`
4	Even numbers: for vs while	Pair programming	`EvenNumbers.java`
5	Loop tracing worksheet	Solo, on paper	Filled-in trace tables
6	Temperature Converter	Module project	`TemperatureConverter.java`

Commit rules: one commit per exercise minimum, and every message says what the code does — for example `Add grade-to-letter logic with if/else if`, never `update`.

Folder layout: put each exercise in its own folder (`module1/division-warmup/`, `module1/grade-calculator/`, …) so your GitHub history stays readable.

Starter — Variable Explorer

This is the independent-practice exercise from Programming Fundamentals 1. Get the starter file one of two ways:

- **From the starter repo:** open [starter repo link], click `VariableExplorer.java`, and copy it into your `claim-academy-[name]-sdai-2026-cohort` project.

- **From this packet:** create a class named `VariableExplorer` in IntelliJ and paste the code below over it.

Fill in every `____` blank, run it, then commit with a descriptive message.

```
public class VariableExplorer {
    public static void main(String[] args) {

        // ---------- Part 1: declare 8 variables ----------
        String firstName = "____";
        String city = "____";

        int age = ____;
        int favoriteNumber = ____;

        double height = ____;      // e.g. your height in meters
        double coffeePrice = ____; // e.g. 4.50

        boolean likesJava = ____;
        boolean isMorningPerson = ____;

        // ---------- Part 2: print each one with a label ----------
        System.out.println("First name: " + firstName);
        System.out.println("City: " + ____);
        System.out.println("Age: " + ____);
        System.out.println("Favorite number: " + ____);
        System.out.println("Height: " + ____);
        System.out.println("Coffee price: " + ____);
        System.out.println("Likes Java: " + ____);
        System.out.println("Morning person: " + ____);

        // ---------- Part 3: one operation per type ----------
        String intro = firstName + " lives in " + ____;   // join two Strings
        System.out.println(intro);

        int intSum = ____ + ____;                         // add two ints
        System.out.println("Age + favorite number = " + intSum);

        double doubleSum = ____ + ____;                   // add two doubles
        System.out.println("Height + coffee price = " + doubleSum);

        if (____) {                                       // check a boolean
            System.out.println("Java fan confirmed!");
```

```
        } else {
            System.out.println("Not a Java fan... yet.");
        }
    }
}
```

- [ ] Done: all blanks filled, the program runs, and it's committed to GitHub.

Exercise 1 — Division & casting warm-up

Predict each result on paper first, then print every expression in Java and compare. Java decides the result type from the operands: `int / int` throws away the remainder.

#	Expression	Your prediction	Java printed	Match?
1	`5 / 2`			
2	`5.0 / 2`			
3	`5 / 2.0`			
4	`(double) 5 / 2`			
5	`(double) (5 / 2)`			
6	`17 % 5`			
7	`-7 / 2`			
8	`(int) 9.99`			
9	`10 / 4 * 4`			
10	`1 + 2 + "3"`			
11	`"1" + 2 + 3`			
12	`7 / 2 == 3`			

Starter code:

```
public class DivisionWarmup {
    public static void main(String[] args) {
        System.out.println("1:  " + (5 / 2));
        System.out.println("2:  " + (5.0 / 2));
        // ... add lines 3 to 12 the same way
    }
}
```

Reflect in two sentences: why do rows 4 and 5 differ, and why do rows 10 and 11 differ?

Exercise 2 — Guided lab: Grade Calculator

You build this with the instructor, one step at a time; don't jump ahead. Commit after every step.

Step 1 — Ask for a grade. Read one number from 0 to 100 with `Scanner`.

```
import java.util.Scanner;

public class GradeCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a grade (0-100): ");
        int grade = input.nextInt();
        System.out.println("You entered " + grade);
    }
}
```

- [ ] Checkpoint: the program echoes the number you typed.

Step 2 — Convert to a letter. Use `if / else if / else`. Order matters: check the highest range first.

```
String letter;
if (grade >= 90) {
    letter = "A";
} else if (grade >= 80) {
    letter = "B";
} else if (grade >= 70) {
    letter = "C";
} else if (grade >= 60) {
    letter = "D";
} else {
    letter = "F";
}
System.out.println("Letter grade: " + letter);
```

- [ ] Checkpoint: test the boundaries 90, 89, 80, 79, 60, 59 and 0. A 90 must print A — that's why it's `>= 90`, not `> 90`.

Step 3 — Allow up to 5 grades. Wrap steps 1–2 in a `for` loop that runs 5 times.

```
for (int i = 1; i <= 5; i++) {
    System.out.print("Grade " + i + " (0-100): ");
    int grade = input.nextInt();
    // letter-grade logic from Step 2
}
```

- [ ] Checkpoint: the loop asks exactly 5 times — not 4, not 6.

Step 4 — Print the average. Keep a running total (the accumulator pattern) outside the loop, then divide as a `double`.

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    // ... read grade, print letter
    total = total + grade;
}
double average = (double) total / 5;
System.out.println("Average: " + average);
```

- [ ] Checkpoint: grades 90, 85, 77, 64, 100 print an average of 83.2, not 83.

Stretch: reject grades below 0 or above 100 and ask again without counting the bad entry.

Exercise 3 — Independent practice: Number Game

The computer picks a secret number from 1 to 10 and you get 3 guesses. This puts `if/else`, a loop and comparison operators to work together.

Requirements

1. Generate a random number from 1 to 10 (use the line below — we'll study it later).
2. Ask the user to guess.
3. Use a loop that allows up to 3 guesses.
4. After each guess, print **Too high**, **Too low** or **Correct!**
5. End with a win message, or a loss message that reveals the number.

The one line you're given:

```
int secret = (int) (Math.random() * 10) + 1;
```

Sample run

```
Guess a number from 1 to 10: 7
Too high.
Guess a number from 1 to 10: 3
Too low.
Guess a number from 1 to 10: 5
Correct! You won in 3 guesses.
```

Hints

- A `for` loop from 1 to 3 counts the guesses for you.
- Leave the loop early with `break` when the guess is correct.
- Use a `boolean won = false;` before the loop so you know which message to print after it.

- [ ] Done: tested a win on guess 1, a win on guess 3, and a loss.
- [ ] Committed `NumberGame.java` with a descriptive message.

Exercise 4 — Pair programming: even numbers, two ways

Both partners print the even numbers from 1 to 20 — one with a `for` loop, one with a `while` loop — then compare.

1. **Partner A writes a `for` loop** that counts from 1 to 20 and prints only the even numbers (hint: `%`).
2. **Partner B reviews** and explains out loud what each line does: where the counter starts, when the loop stops, what changes each pass.
3. **Partner B writes a `while` loop** that does the same task.
4. **Compare together** and write your answers below.

Expected output: `2 4 6 8 10 12 14 16 18 20`

Discussion questions

- Which version was easier to read, and why?
- In the `while` version, what happens if you forget to increase the counter?
- Could you skip `%` entirely? (Try starting at 2 and adding 2 each pass.)
- When would you choose `while` over `for` in a real program?

Exercise 5 — Loop tracing worksheet

The computer runs one line at a time. For each snippet, write every variable's value after each pass, then run the code to check.

Trace A — accumulator

```
int sum = 0;
for (int i = 1; i <= 4; i++) {
    sum = sum + i;
}
System.out.println(sum);
```

Pass	i	sum after the pass
1		
2		
3		
4		

Printed: _____

Trace B — counter with while

```
int n = 20;
int count = 0;
while (n > 1) {
    n = n / 2;
    count++;
}
System.out.println(count);
```

Pass	n after the pass	count after the pass
1		
2		
3		
4		

Printed: _____ (watch the integer division)

Trace C — early exit with break and continue

```
for (int i = 1; i <= 10; i++) {
    if (i % 3 == 0) continue;
    if (i > 7) break;
    System.out.print(i + " ");
}
```

i	Skipped by continue?	Loop ended by break?	Printed
1			
2			
3			
4			
5			
6			
7			

i	Skipped by continue?	Loop ended by break?	Printed
8			

Trace D — do-while

```
int x = 10;
do {
    System.out.println(x);
    x++;
} while (x < 5);
```

How many times does the body run, and why is that different from a `while` loop with the same condition? _____

Exercise 6 — Module project: Temperature Converter

Build a menu-driven program that converts temperatures between Fahrenheit, Celsius and Kelvin, and keeps running until the user chooses to exit.

Requirements

1. Show a menu with three conversions plus an exit option.
2. Repeat the menu in a loop until the user picks Exit.
3. Read a temperature and print the converted result to 2 decimal places.
4. Validate input: if the user types a letter instead of a number, print a friendly message and ask again — the program must not crash.
5. Commit to GitHub with a descriptive message.

Formulas

```
C = (F - 32) \times \frac{5}{9} \qquad F = C \times \frac{9}{5} + 32 \qquad K = C + 273.15
```

Trap: in Java, `5 / 9` is `0`. Write `5.0 / 9` so the math stays in decimals.

Sample run

```
=== Temperature Converter ===
1) Fahrenheit to Celsius
2) Celsius to Fahrenheit
3) Celsius to Kelvin
4) Exit
Choose an option: 1
Enter temperature: abc
That's not a number. Try again: 98.6
98.60 F = 37.00 C

Choose an option: 4
Goodbye!
```

Input-validation pattern

```
while (!input.hasNextDouble()) {
    System.out.print("That's not a number. Try again: ");
    input.next(); // throw away the bad input
}
double temp = input.nextDouble();
```

Suggested rubric

Criterion	What earns full credit	Points
Conversions	All three formulas correct, decimals not truncated	30
Menu loop	Repeats until Exit; invalid menu choice handled	25
Input validation	Letters never crash the program	20
Readability	Clear variable names, a comment on each section	15
Git	Descriptive commit messages, pushed to GitHub	10

Stretch: add Kelvin to Celsius and Fahrenheit to Kelvin, and refuse temperatures below absolute zero.

Quiz prep and submission checklist

The quiz has 12 questions and you pass with 8 correct. Score below 70% and you complete a 30-minute loops exercise before the next module starts.

Topic	Questions	Practice with
Data type from a value	4	Exercise 1
Loop output tracing	4	Exercise 5
Boolean logic evaluation	4	The truth tables in the slides

Before you leave

- [ ] Exercise 1 warm-up table filled in and `DivisionWarmup.java` pushed
- [ ] Grade Calculator steps 1–4 committed
- [ ] Number Game pushed
- [ ] Pair-programming answers written with your partner
- [ ] Loop tracing worksheet filled in
- [ ] Temperature Converter started (due with this module's work)
- [ ] GitHub Foundations exam taken during this module
- [ ] Resume draft sent to your Career Coach

Java syntax cheat sheet

Keep this open while you code — look things up instead of memorizing them.

Types

Type	Holds	Example
`int`	Whole numbers	`int age = 27;`
`double`	Decimals	`double price = 19.99;`
`boolean`	true or false	`boolean isOpen = true;`
`char`	One character, single quotes	`char grade = 'A';`
`String`	Text, double quotes	`String city = "Lagos";`

Operators

Kind	Operators	Watch out for
Arithmetic	`+ - * / %`	`7 / 2` is `3`; use `7.0 / 2` for `3.5`
Update	`+= -= *= /= ++ --`	`x += 5` means `x = x + 5`
Comparison	`== != < > <= >=`	Compare Strings with `.equals()`, not `==`
Logic	`&&` (and), `||` (or), `!` (not)	`&&` needs both sides true
Casting	`(int) 9.99`	Truncates to `9`; it never rounds

Reading input

```
import java.util.Scanner;              // top of the file
Scanner input = new Scanner(System.in);
int n = input.nextInt();               // whole number
double d = input.nextDouble();         // decimal
input.nextLine();                      // clear the leftover Enter
String line = input.nextLine();        // whole line of text
```

Decisions

```
if (grade >= 90) {
    System.out.println("A");
} else if (grade >= 80) {
    System.out.println("B");
} else {
    System.out.println("Keep going");
}
```

Loops

```
for (int i = 1; i <= 5; i++) { ... }   // you know how many times

while (guess != secret) { ... }        // repeat until something changes

do { ... } while (choice != 4);        // runs at least once

break;     // leave the loop now
continue;  // skip to the next pass
```

When something goes wrong

- Red underline before running = compile error: read the message and the line number.
- Crash while running = runtime error: find the exception name and the `(File.java:line)` part.
- Loop acting strange: set a breakpoint, click Debug, press F8 to step, and watch the Variables pane.
- Program never stops: click IntelliJ's red Stop button.